

The amplitude and diagonal corrections for nonattacking bishops

Report 219

4 October 2026

Abstract

Let B_n count placements of n indistinguishable nonattacking bishops on an $n \times n$ board, without identifying placements under board symmetries. For the unique $r \in (1, 2)$ satisfying $r/(1 - e^{-r}) = 2$, we prove that the amplitude recorded numerically in OEIS A238260 is $e^r/(2\pi\sqrt{r^2 - 1})$. We give a self-contained all-fixed-order expansion of OEIS A002465 about Kotěšovec's stated leading equivalent, an exact rational-function coefficient rule, the first two corrections, and the first parity difference $r^3/[4(r + 1)]$ at order n^{-3} . The proof derives the exact formula from the board, controls the moving amplitude uniformly, and proves that the coefficient torus has only one maximum. A parity-preserving inverse is stated with two ceilings, including at exact thresholds. The companion code uses independent counting and symbolic coefficient routes. All remainder and onset claims here are eventual; numerical diagnostics are not certified error bounds.

[write] **Status and trust boundary, added 6 October 2026, batch 109.** This is bundle Report 219 of an external AI-assisted research session, filed in ProveIt as the single-source research report **a002465-nonattacking-bishops** (provenance, the sources as the write read them, what was checked, relation to the repository, the collected non-claims and the reading conventions are in Section 1.1). It is unrefereed and not formalized. Its author line reads “Report 219” and its PDF author field “Research report”: it names no person or tool. *What rests on what.* Theorems 1.1 and 7.1 have conventional proofs: the board derivation of Section 2, the uniform amplitude Lemma 3.1, the unique torus maximum and the domination argument of Section 4, and Gaussian moment algebra. No proof uses a computation. The explicit rational functions $c_1, c_2, c_3^{(p)}$ of (4) and Appendix A are outputs of the finite rule (27), computed by the shipped programs and compared there with an independent affine route; only the parity difference (3) is also derived by hand (Section 6). The decimal table of Section 1 and every numerical residual are diagnostics. The exact enumeration, the exponential base q and the leading equivalent are prior (Perott, Arshon, Kotěšovec, Santos); what is new relative to the sources the manuscript inspected is the closed form of the amplitude b (OEIS A238260) and the corrections. All remainder constants and onsets are existential. The write adds Remark 1.2 (the OEIS entries quoted, the amplitude against all 105 printed digits of A238260, and a Lambert form of b), Proposition 2.2 (the counts at least double from $n = 1$ on), Remark 7.2 (the inverses of Section 7 against the transseries volume), dated notes, and two questions in Section 9. No statement, proof or number of the source was changed.

1 Main result and its scope

Write $B_0 = 1$. For $n \geq 1$, a placement is a subset of n cells, and two bishops attack if their cells share either diagonal. Define

$$\frac{r}{1 - e^{-r}} = 2, \quad q = \frac{e^r}{r} = \frac{2}{r(2 - r)}, \quad b = \frac{e^r}{2\pi\sqrt{r^2 - 1}}, \quad 1 < r < 2. \quad (1)$$

The function on the left is strictly increasing on $(0, \infty)$: its derivative has positive numerator $1 - (1+r)e^{-r}$. Its limit at zero is 1, and its values at 1 and 2 lie on opposite sides of 2. Thus r exists and is unique. The rearranged equation $e^r(2-r) = 2$ also has the inadmissible root 0; throughout, r means the positive root in $(1, 2)$.

Theorem 1.1. *For each fixed integer $J \geq 0$, there are explicitly computable rational functions $c_j^{(p)}(r)$, with $p \in \{0, 1\}$ and $c_0^{(p)} = 1$, such that*

$$B_n = b\Gamma(n)q^n \left(\sum_{j=0}^J \frac{c_j^{(n \bmod 2)}(r)}{n^j} + O_J(n^{-J-1}) \right). \quad (2)$$

The error constant can be chosen for both parities at once. The first two corrections agree across parity, while

$$c_3^{(1)} - c_3^{(0)} = \frac{r^3}{4(r+1)} > 0. \quad (3)$$

A finite exact rule for every fixed $c_j^{(p)}$ is given in Section 5.

For concise explicit formulas, set

$$\begin{aligned} C_1(r) &= 3r^6 - 4r^5 + 7r^4 - 4r^3 - 21r^2 + 12r + 12, \\ C_2(r) &= 9r^{13} - 24r^{12} - 14r^{11} + 280r^{10} - 69r^9 - 1292r^8 \\ &\quad + 478r^7 + 2388r^6 - 495r^5 - 1788r^4 - 816r^3 + 288r^2 + 1152r + 288. \end{aligned}$$

Then

$$c_1 = -\frac{rC_1(r)}{12(r-1)^3(r+1)^2}, \quad c_2 = \frac{rC_2(r)}{288(r-1)^6(r+1)^4}. \quad (4)$$

Appendix A gives the full third coefficient. The companion machine-readable data contain both parity formulas and the amplitude polynomials.

Quantity	Numerical value, rounded
r	1.5936242600400401
q	3.0882773047417402
b	0.6312668788741155
c_1	-1.3945475876486183
c_2	2.3422725603922019
$c_3^{(0)}$	-4.2624100995985707
$c_3^{(1)}$	-3.8722968090405358

These decimal values are diagnostics, not interval certificates.

Relation to previous work. Kotěšovec states the leading equivalent $B_n \sim bq^n(n-1)!$ as a theorem on page 249 of his book, with symbolic q and numerical b [3]. The exact enumeration and exponential base are prior results. OEIS A238260 records the amplitude numerically and by its limit [2]; A002465 records the counting sequence and leading asymptotic [1]. The purpose here is an explicit evaluation of that amplitude and controlled diagonal corrections. The inspected book

passage, the OEIS entries, and the close sources discussed in Section 9 do not supply these symbolic corrections. This is a bounded comparison, not a claim of global historical priority. In particular, the original Arshon and Robinson full texts have not been inspected for this report.

Theorem 1.1 is an expansion for each *fixed* order. It asserts neither convergence of the infinite formal series nor bounds as J grows. The proof gives eventual error constants but does not evaluate a numerical remainder constant or a finite usable onset. Standard saddle-point methods and fixed-piece parity phenomena are prior; the regime here is the diagonal one in which the number of pieces equals the board size.

Remark 1.2 ([write] The OEIS entries, quoted, and the amplitude, 6 October 2026). The entries were read on 6 October 2026 in the OEIS internal format; the quotations are verbatim.

(a) A002465 [1] (revision #95, 19 July 2026) is named

Number of ways to place n nonattacking bishops on an $n \times n$ board.

with offset 0 (the term $a(0) = 1$ was prepended on 1 December 2024, the convention $B_0 = 1$ used here). Its data are 1, 1, 4, 26, 260, 3368, 53744, 1022320, 22522960, 565532992, 15915225216, 496911749920, 17029582652416, ..., through $n = 17$, and its formulas read

Asymptotic: $a(n)/(n-1)! \sim 0.631266 * 3.08827^n$. - _Vaclav Kotesovec_, Mar 23 2011

The second constant is $2/(z*(2-z)) = 3.0882773047417401791158400820254\dots$, where z is the root $z=1.593624260040\dots$ of the equation $\exp(z)*(2-z)=2$. - _Vaclav Kotesovec_, May 27 2011

For constants see A238258 and A238260. - _Vaclav Kotesovec_, Feb 21 2014

so $z = r$ and the second constant is q of (1). The values printed in Section 8 agree with the data. The write compared all 376 terms of the entry's b-file (Kotěšovec, $0 \leq n \leq 375$) with the Stirling convolution (10), and the terms $n \leq 40$ and $n = 100, 101, 200, 201, 374, 375$ with the rook recurrences (6)–(8), in exact integer arithmetic with its own code: all agree.

(b) A238260 [2] (revision #6, 12 September 2015) is named

Decimal expansion of a multiplicative constant related to A002465.

and its only formula is

Equals $\lim_{n \rightarrow \infty} A002465(n) / ((n-1)! * A238258^n)$.

with the example “0.63126687887411546797...” and a link to page 249 of Kotěšovec's book. Its data print 105 digits. Since $\Gamma(n) = (n-1)!$ and A238258 is q (item (c)), the case $J = 0$ of Theorem 1.1 says exactly that this limit is $b = e^r / (2\pi\sqrt{r^2 - 1})$: the source's closed form is the value of a constant that the entry gives only by its limit and its digits. The write evaluated b at 150 significant digits (mpmath); *all 105 printed digits agree*, and the first 110 decimals are

$b = 0.6312668788\ 7411546797\ 0048232579\ 7068795561\ 5469051446\ 1140892006$
 $9734050854\ 1503766170\ 8560400850\ 1761109335\ 4635579420\dots,$

truncated, not rounded (the entry ends at the digit 5 before 79420). This is a closed form for the entry, not a correction of it. Nothing was submitted to the OEIS.

(c) A238258 [9] (revision #10, 1 March 2021) gives q as

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lim n->infinity (A002465(n)/(n-1)!)^(1/n) and -2 / (LambertW(-2*exp(-2)) * (2 +
LambertW(-2*exp(-2))));
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all 105 printed digits agree with the write's value of e^r/r . The saddle r itself is A256500 [10] (revision #36), “Decimal expansion of the positive solution to $x = 2*(1-\exp(-x))$.”, which is (1) rewritten. A *Lambert form of b* (added by the write). Put $w_0 = W_0(-2e^{-2}) = -0.40637573995995990767\dots$ (A226775 [11]). Then

$$r = 2 + w_0, \quad e^r = -\frac{2}{w_0}, \quad q = -\frac{2}{w_0(2 + w_0)}, \quad b = -\frac{1}{\pi w_0 \sqrt{(1 + w_0)(3 + w_0)}}.$$

Proof. With $s = r - 2$ the equation $e^r(2 - r) = 2$ reads $se^s = -2e^{-2}$. Since $1 < r < 2$, $s \in (-1, 0)$, where $\sigma \mapsto \sigma e^\sigma$ is injective with inverse W_0 ; so $s = w_0$. (The other real solution, $s = -2 = W_{-1}(-2e^{-2})$, is the inadmissible root $r = 0$ of Section 1.) Then $e^r = 2/(2 - r) = -2/w_0$, $q = 2/(r(2 - r))$ and $r^2 - 1 = (1 + w_0)(3 + w_0)$ give the displayed forms; $b > 0$ because $w_0 < 0$. The write checked the three identities at 150 digits.

(d) Kotěšovec's asymptotic in (a) is the leading term of Theorem 1.1, with b and q truncated to six digits (0.631266, not the rounded 0.631267); the source credits it as prior (its “Relation to previous work”).

1.1 Provenance, sources and reading conventions

[write] **Added 6 October 2026, batch 109.** *Provenance.* This report is Report 219 of the session bundle that ProvelT's docs/incoming intake processed as batch 109: the archive `Report219-reproducibility.zip` (440,564 bytes, SHA-256 `a0561728f630...ceaa454cc7`, 27 files under the wrapper directory `Report219/`), arrival commit `60f54ea06`, placed in `f7e9e5c2f`. The text is the delivered `Report219.tex` (534 lines, dated 4 October 2026), shipped as `article.tex`; the delivered programs and data are shipped under `code/` and `data/`, and the delivered `code/README.md` as `code-README.md` (the report README maps every name). The delivered 13-page PDF and the checksum manifest `MANIFEST.json` (26 entries, verified at placement) are not shipped, and the delivered README is replaced by the report README; all three are retrievable from the arrival commit. The package records no ProvelT commit and names no repository path. It carries no “prepared for private review” line and no personal data. It is a single-source report, so no merge choice was made. The write prefixed the 49 delivered labels with `bsh:` and updated the 32 references to them; it added a label to Section 1, this subsection, the notes marked [write], Remarks 1.2 and 7.2, Proposition 2.2, two questions in Section 9 and three bibliography entries. The added statements are the last of their sections, the added subsection follows the last delivered text of Section 1, and the added displays are unnumbered, so every theorem, section and equation of the source keeps its delivered number.

The sources, as the write read them (6 October 2026).

- OEIS A002465 with its b-file, A238260 and A238258, and the constants A256500 and A226775, in the internal format (Remark 1.2).
- Santos [5]: Theorem 4 gives the recurrences (6) and (7) for the rook numbers of the white and black boards in exactly this form (its (1) and (2)), as the source says; Theorem 3 is the Stirling formula, “first obtained by Arshon [...] and then refined by Kotěšovec” (book, p. 248), for which Santos provides “an alternative proof”; Theorem 6 is the convolution (8) of the two colours. Santos attributes the exact period 2 for every fixed $k \geq 3$ to Chaiken, Hanusa and Zaslavsky ([7], Theorem 1.1), the period-two theorem the source cites.
- The transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`: `p0:def:core`, `p0:thm:lambert-core`,

p0:prop:factorial-core, p0:thm:core-reversion, p0:rem:core-instances,
p0:def:three-inverses and p0:thm:staircase, read for Remark 7.2.

- Not read by the write: Kotěšovec’s book [3] (the address the source gives returned “404” on 6 October 2026, behind a certificate chain this machine does not trust), so the source’s account of pages 242–253 is reported, not confirmed; Perott [4]; Chaiken, Hanusa and Zaslavsky [6, 7]; Finch [8]; Arshon’s 1936 paper (linked from A002465) and Robinson’s 1976 paper, which the source did not read either.

What was checked at intake. The batch-109 dossier read the manuscript in full and found no error. Independently of the delivered code it enumerated the placements cell by cell for $n \leq 7$ ($B_0, \dots, B_7$ as printed), recomputed B_{10} from its own rook recurrence, and found the scaled residual $n^3(B_n/(b\Gamma(n)q^n) - 1 - c_1/n - c_2/n^2)$ equal to $-4.2222, -3.8389, -4.2422, -3.8555$ at $n = 200, 201, 400, 401$, against $c_3^{(0)} = -4.2624$ and $c_3^{(1)} = -3.8723$; it reran the delivered suite on copies (note at the end of Section 8). The write rechecked every proof line by line and found no error: the monotonicity of $r/(1 - e^{-r})$ and the bracket $(1, 2)$; the board recurrences against the row lengths $n - 1 + \pi(n)$ and $n - \pi(n)$; the factorial identity in the proof of Proposition 2.1; the radius, the bound $(3a^2 - a)/(8(a + 1)^2)$ and the Cauchy tail (17) of Lemma 3.1; the saddle identity $F(r, r) = re^r$, the Hessian (20) and the normalization $b\Gamma(n)q^n$; and the parity computation of Section 6 term by term, including $\mathbb{E}(U - V)^2 = 2/(r + 1)$. With its own code (SymPy 1.14.0, mpmath 1.3.0) it checked that the printed numerators C_1, C_2, C_3 give the coefficients of the shipped data file, that the odd and even third coefficients there differ by exactly $r^3/(4(r + 1))$ and that c_1, c_2 are parity-free; that all seven decimals of the table in Section 1 are correctly rounded; that the amplitude polynomials (18) fit $H_{n,n/2+\delta}$ at $n = 4000, 4001$ (after subtracting them, n^4 times the remainder is $0.065, -0.383, 0.543$ at $x = 1.3, \delta = 0, \frac{1}{2}, -\frac{1}{2}$); the OEIS comparisons of Remark 1.2; the scaled residual above at $n = 300, 301, 374, 375$ ($-4.2355, -3.8499, -4.2408, -3.8543$) from the b-file; and, for the angular-only order-4 receipt, that the n^4 -scaled residuals at $n = 300, 374$ and $301, 375$, extrapolated linearly in $1/n$, give 8.1154 and 6.7763 against the receipt’s $c_4^{(0)} = 8.11568$ and $c_4^{(1)} = 6.77647$. Finite and floating computations are observations, not certificates.

Relation to the repository. Before batch 109 no file of the repository named A002465, A238260 or A238258, and no report counted nonattacking bishops. No statement of this report is formalized in Lean or Rocq, and its place in the collection confers no formal status. The nearest report is `a201513-sparse-chess-placements`, in the same directory `SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/` as this one. It treats the same regime of n pieces on the $n \times n$ board, for kings, knights, wazirs and ferses. It neither contains nor implies the present results: its general theorem (`scp:thm:general`) assumes a finite symmetric move set S , and a bishop is a rider, outside that hypothesis; the growth differs ($n^{2n}e^{\alpha_2}/n!$ there, $b\Gamma(n)q^n$ here), and so does the method (cluster expansion of an independence polynomial there, a saddle point on an exact coefficient formula here). Its B_n is the carrier $e^{-9/2}n^{2n}/n!$, not this count. The batch-109 report `a137432-maximum-density-kings` treats kings at maximum density, a different regime. The inverses of Section 7 are compared with the transseries volume in Remark 7.2.

[write] What is not claimed, collected from the source. Theorem 1.1 holds for each fixed order; neither convergence of the formal series nor bounds uniform in J are asserted, and every remainder constant and onset is existential, with no numerical value. The exact enumeration, the exponential base and the leading equivalent are prior; the source comparison is “a bounded comparison, not a claim of global historical priority”, and the original Arshon and Robinson texts were not inspected. Standard saddle-point methods and fixed-piece parity phenomena are prior; the period-two theorem of Chaiken, Hanusa and Zaslavsky for fixed $k \geq 3$ “does not assert the diagonal expansion when $k = n$ ”. The decimal table and all numerical residuals are diagnostics, not interval certificates. The cosine interpolation (29) is a chosen model, “not a canonical interpolation”; no single-ceiling formula and no uniform real approximation $N(y) = t + o(1)$ follow, and no effective D_J or onset is given. Agreement of finite exact routes checks implementations and conventions, not the theorem; the order-4 receipt is angular-only, with no independent order-4 comparison. PDF byte stability refers to the pinned toolchain. The write adds: its independent checks above are floating or finite; Remark 1.2 gives a closed form for an OEIS constant and corrects nothing; Remark 7.2 claims no novelty for any inversion.

Reading conventions (letters the source reuses, with the tempting false readings; no symbol of the source was renamed; symbols of the transseries volume carry the subscript “vol” in Remark 7.2).

Symbol	Meaning here (where)	Not to be read as
B_n, b, b_n	the bishop count; the amplitude (1); the appended row length (Section 2); b is also a summation index in (10)	a201513’s carrier B_n ; each other
W, K	the white and black Ferrers boards (5), rook numbers $W(n, k)$, $K(n, k)$; K is also the Hessian (21); W is also Lambert’s function in (33)	each other
h, g	$\lceil n/2 \rceil$, $\lfloor n/2 \rfloor$ (Section 2); $n/2 + \delta$ (Section 3); the inverse shift in (33)	the volume’s h_{vol}
t, T, T_m	the variable of $\mathcal{Q}_a(t)$ (Lemma 3.1); the Lambert centre and T of (35) (Section 7); Touchard polynomials (15)	each other
$L_s, L; A_{j,k}, A$	polynomials (13), $\log y$; amplitudes (25), $\log(qt)$	each other
ρ_a, ρ, ρ_J	Cauchy radius (Lemma 3.1); the local ball (Section 4); the inverse radius (Theorem 7.1)	each other
$M_\delta, M_{a,b}, M_J$	amplitude series (26); Gaussian moments; a moment exponent	each other
Ψ, Φ_J, G	the torus phase (19); the inverse model (30); the phase exponential (24)	each other
r, r_k, e_k, d	the saddle; k -matchings and elementary symmetric polynomials in (36); the diagonal label $i - j$ (d in the shipped JSON is δ)	each other
$\pi(n), \delta, \varepsilon$	$n \bmod 2$; the half-shift $\in \{0, \pm \frac{1}{2}\}$; $n^{-1/2}$	the number π ; a small positive constant
$N(y)$	$\min\{n \geq n_0 : B_n \geq y\}$	the staircase N_* of <code>p0: def: three-inverses</code> (equal, Remark 7.2)

2 An exact formula derived from the board

Let $S(a, m)$ denote a Stirling number of the second kind, with $S(0, 0) = 1$ and $S(a, m) = 0$ when $m < 0$ or $m > a$. Falling factorials are denoted by $(h)_{\underline{a}}$. All sums below have nonnegative upper Stirling arguments.

2.1 The two Ferrers boards

Represent a cell (i, j) , $1 \leq i, j \leq n$, by the two diagonal labels $d = i - j$ and $s = i + j$. A set of nonattacking bishops is exactly a matching between the two families of diagonals. The parity of s (equivalently of d) splits this graph into its two square colors.

For a fixed d , the possible s form the interval

$$|d| + 2, |d| + 4, \dots, 2n - |d|.$$

Within one color these intervals have a common center and are nested as $|d|$ decreases. Ordering the columns by successive central pairs makes each neighborhood an initial segment; order the rows by increasing length. Up to exchanging the two colors, and adding one empty row to the shorter

list, the two row-length lists are the first n entries of

$$W : 1, 1, 3, 3, 5, 5, \dots, \quad K : 0, 2, 2, 4, 4, 6, \dots \quad (5)$$

Indeed the diagonal lengths are $n - |d|$, each noncentral length occurring twice, and the central length n once. Separating their odd and even lengths gives exactly (5). The nested neighborhoods make these Ferrers boards. For example, at $n = 4$ their row lengths are $(1, 1, 3, 3)$ and $(0, 2, 2, 4)$.

Let $W(n, k)$ and $K(n, k)$ count k nonattacking rooks on these boards, and write $\pi(n) = n \bmod 2$. On appending a longest row of length b_n , a placement using that row is obtained from a $(k - 1)$ -rook placement on the preceding board. Its $k - 1$ occupied columns all lie in the new row, leaving $b_n - k + 1$ choices. Hence

$$W(n, k) = W(n - 1, k) + (n - k + \pi(n))W(n - 1, k - 1), \quad (6)$$

$$K(n, k) = K(n - 1, k) + (n - k + 1 - \pi(n))K(n - 1, k - 1). \quad (7)$$

Here $W(0, 0) = K(0, 0) = 1$, out-of-range piece counts are zero, and the recurrences are applied for $0 \leq k \leq n$. These are the exact recurrences also given by Santos [5]. The two colors do not interact, so

$$B_n = \sum_{k=0}^n W(n, k)K(n, n - k). \quad (8)$$

2.2 Stirling forms and coefficient extraction

Set $h = \lceil n/2 \rceil$, $g = \lfloor n/2 \rfloor$. To solve (6), put

$$A(n, m) = \sum_{a=0}^h \binom{h}{a} S(n - a, m).$$

The Stirling recurrence and Pascal's identity give

$$A(n, m) = A(n - 1, m - 1) + (m + \pi(n))A(n - 1, m). \quad (9)$$

For an even step, h does not change, and the Stirling recurrence alone proves this. For an odd step, writing $\binom{h}{a} = \binom{h-1}{a} + \binom{h-1}{a-1}$ gives one additional copy of $A(n - 1, m)$. The same calculation with $\lfloor n/2 \rfloor$ gives the coefficient $m + 1 - \pi(n)$. The initial values agree, so

$$W(n, k) = \sum_{a=0}^h \binom{h}{a} S(n - a, n - k), \quad K(n, k) = \sum_{a=0}^g \binom{g}{a} S(n - a, n - k).$$

Inserting these into (8) proves, for all $n \geq 1$,

$$B_n = \sum_{m=0}^n \left(\sum_{a=0}^h \binom{h}{a} S(n - a, m) \right) \left(\sum_{b=0}^g \binom{g}{b} S(n - b, n - m) \right). \quad (10)$$

The endpoint terms vanish for $n > 1$; the formula gives $B_1 = 1$. This is a reindexing of the prior Arshon–Kotěšovec form, for which Santos supplies an alternative proof [5]. The derivation above also fixes all conventions without assuming that formula.

Define

$$H_{n,h}(x) = \sum_{a=0}^h \frac{(h)_a}{(n)_a} \frac{x^a}{a!}, \quad F(x, y) = e^x + e^y - 2. \quad (11)$$

Proposition 2.1. *For every $n \geq 1$,*

$$B_n = n![x^n y^n] F(x, y)^n H_{n,h}(x) H_{n,g}(y). \quad (12)$$

Proof. Expand F^n by the binomial theorem and use $[x^N](e^x - 1)^m = m!S(N, m)/N!$. Thus

$$[x^{n-a} y^{n-b}] F^n = \frac{n!}{(n-a)!(n-b)!} \sum_{m=0}^n S(n-a, m) S(n-b, n-m).$$

The two H coefficients and the exterior $n!$ multiply its factorial factor to

$$n! \frac{(h)_a}{(n)_a} \frac{(g)_b}{(n)_b} \frac{n!}{(n-a)!(n-b)!} = \binom{h}{a} \binom{g}{b}.$$

Equation (10) now proves the claim. $\square$

Proposition 2.2 ([write] The counts at least double, 6 October 2026). *For every $n \geq 1$, $B_{n+1} \geq 2B_n$. In particular $(B_n)_{n \geq 1}$ is positive and strictly increasing, and the onset n_0 of Section 7 may be taken to be 1.*

Proof. All terms of (6) and (7) are nonnegative, so dropping one term of each recurrence at step $n+1$ gives, for $0 \leq k \leq n$ and $0 \leq j \leq n$,

$$W(n+1, k) \geq W(n, k), \quad W(n+1, k+1) \geq (n-k+\pi(n+1))W(n, k),$$

$$K(n+1, j) \geq K(n, j), \quad K(n+1, j+1) \geq (n-j+1-\pi(n+1))K(n, j).$$

Two vanishing facts are used: $K(n, n) = 0$ for $n \geq 1$, because the first row of the black board is empty and n rooks on n rows need every row; and $W(n, n) = 0$ for $n \geq 2$, because the first two rows of the white board have one cell each, in the same column. Apply (8) at $n+1$ and keep only some of its nonnegative terms.

If n is odd, $\pi(n+1) = 0$, and pairing $W(n+1, k)$ with $K(n+1, (n-k)+1)$ gives

$$B_{n+1} \geq \sum_{k=0}^n W(n, k) (k+1) K(n, n-k).$$

The term $k=0$ is $K(n, n) = 0$, and every other factor $k+1$ is at least 2.

If n is even, then $n \geq 2$, $\pi(n+1) = 1$, and pairing $W(n+1, k+1)$ with $K(n+1, n-k)$ gives

$$B_{n+1} \geq \sum_{k=0}^n (n-k+1) W(n, k) K(n, n-k).$$

The term $k=n$ is $W(n, n)K(n, 0) = 0$, and every other factor $n-k+1$ is at least 2. In both cases $B_{n+1} \geq 2 \sum_k W(n, k) K(n, n-k) = 2B_n$. Since $B_1 = 1$, induction gives $B_n \geq 2^{n-1} > 0$ and strict increase for $n \geq 1$. (The b-file of A002465 has $B_{n+1}/B_n \geq 4$ for $1 \leq n \leq 374$; $B_1 = B_0$.) $\square$

3 A uniform expansion for the moving amplitude

For signed $\delta \in \{-\frac{1}{2}, 0, \frac{1}{2}\}$ and admissible n , write $h = n/2 + \delta$. For even n the two factors use $\delta = 0$; for odd n they use $\delta = \frac{1}{2}$ and $-\frac{1}{2}$.

For each integer $s \geq 1$ define the polynomial

$$L_s(a, \delta) = \frac{1}{s} \sum_{k=0}^{a-1} ((-1)^{s+1} (2\delta - 2k)^s + k^s). \quad (13)$$

The finite sum, first defined for integer $a \geq 0$, has a unique polynomial continuation in a . Set

$$Q_0(a, \delta) = 1, \quad Q_j(a, \delta) = \frac{1}{j} \sum_{s=1}^j s L_s(a, \delta) Q_{j-s}(a, \delta). \quad (14)$$

If $Q_j(a, \delta) = \sum_m q_{jm}(\delta) a^m$, put

$$P_j(x, \delta) = \sum_m q_{jm}(\delta) T_m(x/2), \quad T_m(v) = e^{-v} \sum_{a \geq 0} a^m \frac{v^a}{a!}. \quad (15)$$

Here T_m is the Touchard polynomial and $T_0 = 1$.

Lemma 3.1. *For fixed $R > 0$ and integer $J \geq 0$, uniformly on $|x| \leq R$ and on the admissible signed parameters,*

$$H_{n, n/2+\delta}(x) = e^{x/2} \left(\sum_{j=0}^J P_j(x, \delta) n^{-j} + O_{R,J}(n^{-J-1}) \right). \quad (16)$$

Any fixed number of complex derivatives has the same uniform remainder on a smaller disk.

Proof. For $a \geq 0$, introduce the rational function

$$\mathcal{Q}_a(t) = \prod_{k=0}^{a-1} \frac{1 + (2\delta - 2k)t}{1 - kt}.$$

For $a \leq h$, $(h)_a / (n)_a = 2^{-a} \mathcal{Q}_a(1/n)$. Formal logarithmic expansion gives $\log \mathcal{Q}_a(t) = \sum_{s \geq 1} L_s(a, \delta) t^s$, so its Taylor coefficients are the Q_j of (14). Since $\deg_a L_s \leq s + 1$, induction gives $\deg_a Q_j \leq 2j$.

Here is a remainder estimate with no hidden n dependence. On the disk

$$|t| \leq \rho_a = \frac{1}{8(a+1)^2},$$

all numerator and denominator perturbations have modulus below $1/2$. Applying $|\log(1+z)| \leq 2|z|$ termwise gives

$$|\log \mathcal{Q}_a(t)| \leq 2\rho_a \sum_{k=0}^{a-1} (3k+1) = \frac{3a^2 - a}{8(a+1)^2} < \frac{3}{8}$$

for $a \geq 1$; $a = 0$ is immediate. Hence $|\mathcal{Q}_a(t)| < 2$ there. Cauchy's coefficient estimate and a geometric tail imply, when $1/n \leq \rho_a/2$,

$$\left| \mathcal{Q}_a(1/n) - \sum_{j=0}^J Q_j(a, \delta) n^{-j} \right| \leq 4[8(a+1)^2]^{J+1} n^{-J-1}. \quad (17)$$

Split the defining sum at $a \leq \lfloor \log n \rfloor$. For large n this range lies below h and satisfies the Cauchy condition. On $|x| \leq R$, its total error is at most

$$4 \cdot 8^{J+1} n^{-J-1} \sum_{a \geq 0} (a+1)^{2J+2} \frac{(R/2)^a}{a!},$$

a finite constant times n^{-J-1} . For the remaining terms, the original falling-factor ratio lies in $[0, 1]$; extend it by zero for $a > h$. Its tail is bounded by $\sum_{a > \log n} R^a/a!$, which is $o(n^{-M})$ for every fixed M : Stirling's elementary factorial bound makes its logarithm at most $-(\log n) \log \log n + O(\log n)$. The finitely many polynomially weighted approximating tails have the same property. We may therefore extend their sums to infinity. Equation (15) sums each polynomial against $(x/2)^a/a!$ and proves (16). Cauchy estimates on a larger disk prove the derivative statements. $\square$

The first polynomials, useful both for checking normalization and for parity, are

$$\begin{aligned} P_0 &= 1, \\ P_1 &= \delta x - \frac{x^2}{8}, \\ P_2 &= \frac{\delta^2 x^2}{2} - \frac{\delta x^3}{8} + \frac{x^4}{128} - \frac{x^2}{8}, \\ P_3 &= \frac{\delta^3 x^3}{6} - \frac{\delta^2 x^4}{16} + \frac{\delta^2 x^2}{2} + \frac{\delta x^5}{128} - \frac{7\delta x^3}{24} - \frac{x^6}{3072} + \frac{x^4}{32} - \frac{x^2}{8}. \end{aligned} \tag{18}$$

4 The torus and the complete saddle argument

Use the Cauchy torus $x = re^{i\theta}$, $y = re^{i\phi}$. Since $F(r, r) = re^r$, the local phase is

$$\Psi(\theta, \phi) = \log \frac{F(re^{i\theta}, re^{i\phi})}{F(r, r)} - i(\theta + \phi). \tag{19}$$

Only near $(0, 0)$ is this logarithm needed; there F is nonzero. The saddle equation in each variable is $re^r/F(r, r) = 1$. Direct differentiation gives

$$\Psi(\theta, \phi) = -\frac{1}{2}(r\theta^2 - 2\theta\phi + r\phi^2) + O((|\theta| + |\phi|)^3). \tag{20}$$

Thus

$$K = \begin{pmatrix} r & -1 \\ -1 & r \end{pmatrix}, \quad \Sigma = K^{-1} = \frac{1}{r^2 - 1} \begin{pmatrix} r & 1 \\ 1 & r \end{pmatrix}. \tag{21}$$

The eigenvalues $r - 1$ and $r + 1$ are positive.

4.1 There is only one modulus maximum

The power series of F has positive coefficients at all $(k, 0)$ and $(0, k)$, $k \geq 1$. Equality in

$$|F(re^{i\theta}, re^{i\phi})| \leq F(r, r)$$

requires every phase $e^{ik\theta}$ and $e^{ik\phi}$ to agree. Comparing $k = 1$ and $k = 2$ first forces $e^{i\theta} = e^{i\phi} = 1$. Thus the origin is the unique maximum on the torus. On the compact complement of any fixed neighborhood, continuity supplies $\eta > 0$ such that $|F|/F(r, r) \leq 1 - \eta$. Also

$$|H_{n,h}(re^{i\theta})| \leq e^r$$

because its falling-factor ratios lie in $[0, 1]$. The complementary integral is exponentially small, even relative to the local n^{-1} scale. In particular parity does not arise from an overlooked second saddle.

4.2 The normalized local remainder

We supply the domination argument to justify arbitrary fixed order, rather than just a formal expansion. Put $\varepsilon = n^{-1/2}$, $w = (u, v)$, and $l_2(w) = -w^\top K w / 2$. On a small ball $|\varepsilon w| \leq \rho$, define the analytic continuation at $\varepsilon = 0$ of

$$R(\varepsilon, w) = \frac{\Psi(\varepsilon u, \varepsilon v) - \varepsilon^2 l_2(w)}{\varepsilon^2}.$$

Taylor's integral formula gives $R = O(\varepsilon|w|^3)$. For every fixed derivative order, its ε derivatives are bounded by a fixed polynomial in $|w|$ on this ball; this follows either from the integral formula or the convergent power series of Ψ . Take ρ so small that

$$|e^{R(s,w)}| \leq e^{(r-1)|w|^2/4} \quad (0 \leq s \leq \varepsilon).$$

This is possible since $|R(s, w)| \leq C\rho|w|^2$. Multiplication by $e^{l_2(w)}$ leaves the integrable Gaussian majorant $e^{-(r-1)|w|^2/4}$.

Lemma 3.1 replaces the two moving H factors by a finite sum of analytic amplitudes through n^{-J} . Their remainder is uniformly $O(n^{-J-1})$. On the small ball, $\Re(n\Psi) \leq -c|w|^2$, so its integrated contribution has the same normalized order. Compose each retained analytic amplitude with $re^{i\varepsilon u}$ and $re^{i\varepsilon v}$. Every fixed ε derivative is polynomially bounded in $|w|$. Taylor-expand the product of these amplitudes, the factors ε^{2j} , and e^R through degree $2J + 1$. The integral remainder formula bounds the integrated error by

$$C_J \varepsilon^{2J+2} \int_{\mathbb{R}^2} (1 + |w|)^{M_J} e^{-(r-1)|w|^2/4} dw = O_J(n^{-J-1}).$$

No bound uniform in J is being asserted.

The coefficient of an odd power of ε is odd under $w \mapsto -w$: each phase monomial of degree m accompanies ε^{m-2} , and each amplitude derivative of degree k accompanies ε^{2j+k} . It therefore integrates to zero over the symmetric local ball. Extending each remaining polynomial Gaussian integral to all of $\mathbb{R}^2$ changes it by an exponentially small quantity. This proves the required all-fixed-order error, including the moving-amplitude contribution.

4.3 The leading normalization

From (12), the Cauchy prefactor is $n!q^n/(2\pi)^2$. The local change of variables supplies n^{-1} , the leading amplitude is e^r , and

$$\int_{\mathbb{R}^2} e^{-w^\top K w / 2} dw = \frac{2\pi}{\sqrt{r^2 - 1}}.$$

Consequently

$$B_n \sim \frac{n!q^n e^r}{2\pi n\sqrt{r^2-1}} = b\Gamma(n)q^n.$$

The argument above gives every fixed correction. In particular no Stirling approximation to $n!$ has been used in this normalization, so no additional Stirling correction belongs in the coefficients c_j .

5 A finite exact rule for every coefficient

Let (U, V) be a centered Gaussian pair with covariance (21). All operations in this section may equivalently be read as formal polynomial moment operations. Define

$$f_m(U, V) = \frac{i^m T_m(r)}{r m!} (U^m + V^m), \quad m \geq 1, \quad (22)$$

$$l_1 = f_1, \quad l_m = f_m - \frac{1}{m} \sum_{k=1}^{m-1} k l_k f_{m-k}, \quad m \geq 2. \quad (23)$$

Indeed $1 + \sum f_m \varepsilon^m$ is the Taylor series of $F(re^{i\varepsilon U}, re^{i\varepsilon V})/F(r, r)$. Differentiating its formal logarithm proves (23). The linear term is $l_1 = i(U + V)$, canceled by the coefficient kernel, and l_2 is the quadratic form already included in the Gaussian measure.

Set

$$G(\varepsilon; U, V) = \exp \left(\sum_{m \geq 3} l_m(U, V) \varepsilon^{m-2} \right). \quad (24)$$

For $j, k \geq 0$ put

$$A_{j,k}(\delta) = e^{-r/2} \left[(x \partial_x)^k \{ e^{x/2} P_j(x, \delta) \} \right]_{x=r}, \quad (25)$$

and define

$$M_\delta(\varepsilon; U, V) = \left(\sum_{j,k \geq 0} \varepsilon^{2j+k} \frac{i^k U^k}{k!} A_{j,k}(\delta) \right) \cdot \left(\sum_{j,k \geq 0} \varepsilon^{2j+k} \frac{i^k V^k}{k!} A_{j,k}(-\delta) \right). \quad (26)$$

Then the promised coefficient rule is

$$\boxed{c_J^{(p)}(r) = \mathbb{E}([\varepsilon^{2J}] G(\varepsilon; U, V) M_{p/2}(\varepsilon; U, V)).} \quad (27)$$

For order J , one needs only $m \leq 2J + 2$ in the phase and $2j + k \leq 2J$ in the amplitude. Thus this is a finite algorithm for every fixed order, subject only to computing resources.

For completeness, Gaussian moments can be evaluated without an integration routine. Write $M_{a,b} = \mathbb{E}U^a V^b$. Odd total degree gives zero, $M_{0,0} = 1$, and Gaussian integration by parts gives, for $a \geq 1$,

$$M_{a,b} = (a-1)\Sigma_{11}M_{a-2,b} + b\Sigma_{12}M_{a-1,b-1},$$

with terms of zero multiplier omitted. For $a = 0$ and even $b \geq 2$, use $M_{0,b} = (b-1)\Sigma_{22}M_{0,b-2}$. These rules show that every moment is rational in r . Equations (13)–(27) involve only rational algebra, so every $c_J^{(p)}$ is a rational function. The local proof in Section 4 identifies these formal coefficients with those in Theorem 1.1, completing that proof.

5.1 A distinct symbolic cross check

An independent local coordinate choice is

$$x = r(1 + i\varepsilon U), \quad y = r(1 + i\varepsilon V).$$

Now the coefficients of $F/F(r, r)$ are $i^m r^{m-1} (U^m + V^m)/m!$, and the phase is

$$\log(F/F(r, r)) - \log(1 + i\varepsilon U) - \log(1 + i\varepsilon V).$$

The quadratic form is still l_2 , but the amplitude must include the Jacobian

$$\frac{1}{(1 + i\varepsilon U)(1 + i\varepsilon V)}.$$

Locally these contours are analytic deformations of the angular ones: at a fixed small distance from the saddle the connecting arcs remain in a region with strictly negative real phase. Their integrals are exponentially small. The coefficient expansions must therefore agree.

The companion oracle uses finite powers for logarithms, integer partitions for exponentials, and Newton finite differences of the products $\mathcal{Q}_a$ to construct the amplitude polynomials. It evaluates moments by $U = Z + W$, $V = Z - W$, where Z, W are independent centered Gaussians with variances $1/[2(r-1)]$ and $1/[2(r+1)]$. This differs from both the angular differential operators and the correlated-moment recurrence above. Agreement through order three is a finite exact algebra check, not the analytic justification of the theorem.

6 Why parity first appears at third order

The integral is invariant under $x \leftrightarrow y$. Therefore replace its amplitude by the symmetric part, which eliminates all terms odd in δ . Factoring out $e^{(x+y)/2}$ and using (18), the δ -dependent terms relevant through n^{-3} are

$$\begin{aligned} n^{-1} : & \quad \delta(x - y) \quad \text{before symmetrization,} \\ n^{-2} : & \quad \frac{\delta^2}{2}(x - y)^2, \\ n^{-3} : & \quad \delta^2 \left\{ \frac{x^2 + y^2}{2} - \frac{(x - y)^2(x^2 + y^2)}{16} \right\}. \end{aligned}$$

There is no even positive power of δ at the first order. The second displayed term vanishes to second order at $x = y = r$, so it contributes nothing to c_2 . In local angular coordinates,

$$x - y = i r \varepsilon (U - V) + O(\varepsilon^2).$$

Its first contribution to c_3 is therefore

$$-\frac{\delta^2 r^2}{2} \mathbb{E}(U - V)^2 = -\frac{\delta^2 r^2}{r + 1}, \quad \mathbb{E}(U - V)^2 = \frac{2}{r + 1}.$$

The third-order term contributes its value $\delta^2 r^2$ at the saddle. No phase or leading-amplitude correction can enter this calculation at the same order: the second-order difference already starts two local powers later, and the third-order term uses only its constant local term. Their sum is

$$\frac{\delta^2 r^3}{r + 1}.$$

Substitute $\delta = 1/2$ for odd n and 0 for even n to obtain (3). This small-polynomial derivation verifies the first split without relying on the large polynomial in Appendix A.

7 Inverting the count without losing parity

The leading equivalent implies $B_{n+1}/B_n \sim qn$, hence eventual strict increase. Choose an integer n_0 beyond that onset and define

$$N(y) = \min\{n \geq n_0 : B_n \geq y\}.$$

For $y > \max_{1 \leq n < n_0} B_n$ this equals the natural global threshold $\min\{n \geq 1 : B_n \geq y\}$. Thus excluding the finite prefix changes none of the large- y statements.

Define logarithmic coefficient functions by formal expansion:

$$\log \left(1 + \sum_{j \geq 1} c_j^{(p)} u^j \right) = \sum_{j \geq 1} \ell_j^{(p)} u^j. \quad (28)$$

For example, $\ell_1 = c_1$, $\ell_2 = c_2 - c_1^2/2$, and $\ell_3^{(p)} = c_3^{(p)} - c_1 c_2 + c_1^3/3$. Choose the smooth interpolation

$$\ell_j(x) = \frac{\ell_j^{(0)} + \ell_j^{(1)}}{2} + \frac{\ell_j^{(0)} - \ell_j^{(1)}}{2} \cos(\pi x), \quad (29)$$

$$\Phi_J(x) = \log b + \log \Gamma(x) + x \log q + \sum_{j=1}^J \frac{\ell_j(x)}{x^j}. \quad (30)$$

This is a chosen model preserving the integer parity coefficients, not a canonical interpolation of the exact sequence. Its oscillating derivative begins only at x^{-3} . Using the usual real digamma estimate gives

$$\Phi_J'(x) = \log(qx) + O(1/x) > 0 \quad (31)$$

on a terminal interval. For large y , let $t_J(y)$ be the unique root there of $\Phi_J(t) = \log y$.

Theorem 7.1. *For each fixed $J \geq 0$, a constant $D_J > 0$ exists such that, for sufficiently large y , writing $t = t_J(y)$ and*

$$\rho_J(t) = \frac{D_J}{t^{J+1} \log(qt)},$$

one has the pointwise enclosure

$$\lceil t - \rho_J(t) \rceil \leq N(y) \leq \lceil t + \rho_J(t) \rceil. \quad (32)$$

It remains valid when $y = B_n$ exactly. No effective numerical value of D_J or onset is provided.

Proof. Taking logarithms in Theorem 1.1 gives constants K_J and an eventual bound

$$|\log B_n - \Phi_J(n)| \leq K_J n^{-J-1}.$$

Put $A = \log(qt)$ and choose $D_J > 2^{J+2} K_J$. For large t , $\rho = \rho_J(t) < 1$, and (31) implies $\Phi_J' \geq A/2$ on $[t-2, t+2]$. Define

$$n_- = \lceil t - \rho \rceil - 1, \quad n_+ = \lceil t + \rho \rceil.$$

Both integers lie in that interval, and their distances from t in the appropriate direction are at least ρ . The log-remainder bound at either integer is at most $2^{J+1} K_J t^{-J-1}$, while the mean-value displacement is at least $A\rho/2 = D_J t^{-J-1}/2$, strictly larger. Consequently

$$\log B_{n_-} < \log y < \log B_{n_+}.$$

Eventual discrete monotonicity implies $n_- + 1 \leq N(y) \leq n_+$, which is (32). No separation from integers was assumed. $\square$

Although $\rho_J(t) \rightarrow 0$, its two ceiling values may still differ by one arbitrarily close to a threshold. A universal single-ceiling formula, or a uniform real approximation $N(y) = t + o(1)$ for the integer threshold itself, does not follow. Separate parity-branch roots and parity-adjusted ceilings offer an alternative; the cosine construction is a convenient single-root formulation.

7.1 An explicit Lambert inverse center

Let $L = \log y$, use the principal real Lambert branch, and set

$$w = W(qL/e), \quad t = \frac{L}{w}, \quad A = \log(qt), \quad \gamma_0 = \log(b\sqrt{2\pi}), \quad h = \frac{\frac{1}{2} \log t - \gamma_0}{A}. \quad (33)$$

Thus $t(\log(qt) - 1) = L$ and $h = O(1)$. For any fixed $J \geq 1$, the real root $t_J(y)$ has

$$t_J(y) = t + h + O\left(\frac{1}{t \log t}\right), \quad (34)$$

$$t_J(y) = T + O\left(\frac{1}{t^2 \log t}\right), \quad (35)$$

$$T = t + h - \frac{(h^2 - h)/2 + 1/12 + c_1}{tA}.$$

Indeed Stirling's expansion in (30) gives

$$\Phi_J(x) = x(\log(qx) - 1) - \frac{1}{2} \log x + \gamma_0 + \frac{1/12 + c_1}{x} + O(x^{-2}).$$

At $x = t + h$, its residual after subtracting L is

$$\frac{(h^2 - h)/2 + 1/12 + c_1}{t} + O(t^{-2}).$$

One Newton correction proves (35) using derivative $A + O(1/t)$; omitting it proves (34). These centers yield corresponding eventual two-ceiling enclosures with radii of their stated horizontal error scales. For finer inverse orders, retain (30), including its parity term from $J = 3$ onward.

Remark 7.2 ([write] The inverses against the transseries volume, 6 October 2026). The volume is `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`; its letters $\kappa, d, v, X, L, E, h, f, \Lambda$ are written here with the subscript “vol”. Each statement of this section is placed against it separately.

- (a) *The threshold is the volume's staircase (instance).* By Proposition 2.2, $A_n = B_n$ with $n_1 = 1$ satisfies the hypotheses of `p0:def:three-inverses`, and with $n_0 = 1$ the threshold $N(y)$ is its staircase $N_*(y)$ (for another choice of n_0 the two agree when $y > B_{n_0-1}$, as the source notes). So `p0:thm:staircase(1)` gives $N(y) = \lceil \nu_{\mathcal{A}}(y) \rceil$ for $y \geq 1$ and every admissible interpolation $\mathcal{A}$ of $(B_n)_{n \geq 1}$, $\nu_{\mathcal{A}} = \mathcal{A}^{-1}$. The source does not state this.
- (b) *Theorem 7.1 is an analogue, not an instance.* The model e^{Φ_J} is not an admissible interpolation: $\Phi_J(n)$ equals $\log B_n$ only up to $O(n^{-J-1})$, and the enclosure (32) is proved directly from that bound and monotonicity, around the root $t_J(y)$ of the model, with no separation hypothesis. It shares the conclusion of the separation condition of `p0:thm:staircase(2)`: when the two ceilings coincide, $N(y)$ is their common value; otherwise, since $2\rho_J(t) < 1$ eventually, two consecutive values remain. The source's remark that no universal single-ceiling formula follows is in line with `p0:thm:staircase(5)`. The two-ceiling enclosures about the centres (34) and (35) that the source mentions are analogues in the same sense.

- (c) *The parity-branch alternative.* The “separate parity-branch roots and parity-adjusted ceilings” mentioned after Theorem 7.1 have the shape of `p0:thm:staircase(4)` with $r = 2$: by Proposition 2.2 both parity subsequences are strictly increasing, so for any admissible interpolations of the even and odd subsequences the residue-class formula of (4) holds for large y . What the source does not supply is an enclosure of those per-class inverses; the branch centres of `code/inverse.py` are formal (README of the code). This is Question 7 of Section 9.
- (d) *The Lambert centre is an instance of `p0:prop:factorial-core`.* With $\kappa_{\text{vol}} = 1$, $d_{\text{vol}} = \log q - 1$, $X_{\text{vol}} = t$ and $L_{\text{vol}} = L$, the core $X(\kappa \log X + d)$ is $t(\log(qt) - 1)$, the volume’s $v_{\text{vol}} = W_0(Le^{d_{\text{vol}}}) = W_0(qL/e) = w$, and $X_{\text{vol}} = \exp(v_{\text{vol}} - d_{\text{vol}}) = e^w e/q = L/w = t$ (because $we^w = qL/e$). For $y > 1$ (so $L > 0$) the proposition makes t the unique solution of $t(\log(qt) - 1) = L$ with $\log(qt) > 1$, on the principal branch the source uses, and its core slope is $\log(qt) = A = w + 1$. The linear-logarithmic core of `p0:thm:lambert-core` does not apply.
- (e) *The centres (34) and (35) are a formal instance of `p0:thm:core-reversion` after the change of variable $x = t(1 + E)$.* Put $\tau = 1/t$. Inserting $x = t(1 + E)$ in the Stirling form of Φ_J displayed above and using $t(A - 1) = L$ gives $\Phi_J(x) - L = t[\Lambda_{\text{vol}}E + h_{\text{vol}}(E) + f_{\text{vol}}(\tau, E)]$ with $\Lambda_{\text{vol}} = A$, $h_{\text{vol}}(E) = (1 + E)\log(1 + E) - E$ (the h that `p0:rem:core-instances` names for the factorial core) and

$$f_{\text{vol}}(\tau, E) = \tau \left(-\frac{1}{2} \log(1 + E) - Ah \right) + \tau^2 \left(\frac{1}{12} + c_1 \right) (1 + E)^{-1} + \dots,$$

where $Ah = \frac{1}{2} \log t - \gamma_0$ (in this item E is the volume’s unknown $E_{\text{vol}} = \sum_n e_n^{\text{vol}} \tau^n$, not the expectation $\mathbb{E}$). The triangular recurrence of the theorem, $e_n^{\text{vol}} = -\Lambda_{\text{vol}}^{-1}[\tau^n](h_{\text{vol}}(E_{<n}) + f_{\text{vol}}(\tau, E_{<n}))$, gives $e_1^{\text{vol}} = h$ and $e_2^{\text{vol}} = -((h^2 - h)/2 + 1/12 + c_1)/A$, so $x = t + e_1^{\text{vol}} + e_2^{\text{vol}}/t + \dots$ reproduces $t + h$ and T exactly (the write checked both coefficients with SymPy). The instance is only formal: the coefficient ring must contain A^{-1} and $\log t$, which vary with t , so the volume’s series is not by itself an expansion in a fixed scale; the remainders in (34)–(35) are the source’s own Newton argument. Moreover the cosine term (29) first enters at x^{-3} , through $\cos(\pi(1 + E)/\tau)$, which is not a power series in (τ, E) ; the formal instance therefore covers only truncations that stop before it, as the two displayed centres do.

No novelty is claimed for any of these inversions.

8 Exact checks and reproducibility

The accompanying archive is self-contained and contains this TeX source, its PDF, exact coefficient data, public Python programs, build instructions, and deterministic receipts. Third-party papers are linked in the bibliography and are not redistributed. Python’s exact integers and SymPy rational functions are used for the algebra; optional decimal diagnostics use mpmath. Input checks raise explicit exceptions, so optimized Python does not remove them.

The public verification distinguishes four logically separate questions.

1. **Board conventions.** Literal increasing-cell-subset enumeration checks every piece count through $k = n$ on boards $n \leq 5$. A separate rook inclusion–exclusion count on the actual diagonal graph checks larger small boards.
2. **Exact identities.** The board counts, the two Santos recurrences, the Stirling convolution, and direct rational bivariate coefficient extraction check the exact normalization. Selected large even and odd values through 600 compare the recurrence and Stirling routes.

3. **Finite symbolic algebra.** The angular and affine-contour engines independently reconstruct c_0 through c_3 at both parities and compare exact rational functions. Formal logarithm and inverse residual checks retain the first parity term.
4. **Diagnostic behavior.** Optional high-precision evaluation compares exact counts with truncated expansions and parity-aware real inverse models. This is a numerical check only; it supplies no certified asymptotic remainder or onset.

For clarity, the independent inclusion–exclusion formula has a short proof. In a bipartite graph with m column vertices let $d_i(S)$ be the number of neighbors of row i in S . If r_k counts k -matchings and e_k is an elementary symmetric polynomial, then

$$r_k = \sum_{\substack{S \subseteq [m] \\ |S| \leq k}} (-1)^{k-|S|} \binom{m-|S|}{k-|S|} e_k(d_1(S), \dots, d_l(S)). \quad (36)$$

For each fixed set of k rows and k columns, inclusion–exclusion removes functions omitting a chosen column. Surjectivity is then bijectivity. Sum first over the surviving subset S ; there are $\binom{m-|S|}{k-|S|}$ column supersets. Finally summing the products of row degrees over k rows gives e_k . This count uses the actual board graph and does not presume the Ferrers or Stirling formulas.

The first values, starting at $n = 0$, are

$$1, 1, 4, 26, 260, 3368, 53744, 1022320, 22522960,$$

and $B_{10} = 15915225216$, $B_{12} = 17029582652416$. Agreement on a finite range checks implementations and conventions; Sections 2–6 prove the theorem independently of those tests. The archive README specifies the tested ranges, commands, dependency versions, receipt fields, and the resource limits of higher symbolic orders. Rebuilding from the actual archive in fresh directories, both normally and with `python -O`, is the reproducibility target. PDF byte stability refers to the pinned build toolchain, not an assertion that unrelated TeX installations produce identical bytes.

[write] **The archive in the repository and the reruns, 6 October 2026.** In ProveIt the archive is not shipped as such: the TeX source is `article.tex` (labels prefixed, [write] additions), the programs are in `code/`, the coefficient data, requirement pins, toolchain record and reference receipts in `data/`, and the delivered PDF and `MANIFEST.json` are not shipped (the report README maps every name and gives the retrieval from the arrival commit). The delivered programs read their data from beside themselves, so they run only in the delivered layout, rebuilt from the shipped files or re-extracted from the archive. At placement the batch-109 dossier ran, on copies, `verify_package.py` (26 manifest entries PASS), `reproduce.pyall--with-diagnostics` (SymPy 1.14.0, mpmath 1.3.0; PASS in about 36 s, normal and `-O` outputs identical, all five receipts equal to the references up to line endings), `test_contracts.py` (receipts equal to the references) and the order-4 run with `-skip-oracle` (about 2 min, equal to the reference). The write rebuilt the delivered layout from the shipped files (every rebuilt file byte-identical to the archive) and reran `reproduce.pyall--with-diagnostics` (PASS; `compare_receipts.py` finds identical semantic JSON in all five receipts; byte equality holds after removing the carriage returns that Windows text-mode writes add) and `test_contracts.py` (PASS; receipts equal after the same removal). The PDF and archive builders were not run. The order-4 receipt is angular-only: its coefficients have no independent exact comparison (Question 6); the write’s numerical comparison is in Section 1.1.

9 Source comparison and further questions

The source comparison is intentionally narrow and explicit. Perott’s 1883 paper splits colors and develops exact enumeration, including the eight-bishop count [4]. The inspected pages 242–253 of

Kotěšovec’s sixth edition include the exact Stirling formula and the stated leading asymptotic, but display the amplitude numerically [3]. Santos gives modern exact recurrences and closed forms and studies coefficients in the fixed-piece regime [5]. Chaiken, Hanusa, and Zaslavsky record the exact bishop formula and investigate quasipolynomial parity for a fixed number of pieces [6, 7]. Their period-two theorem in board size for fixed $k \geq 3$ does not assert the diagonal expansion when $k = n$. Finch’s inspected bishop references point back to Kotěšovec and A002465 [8].

The original Arshon paper is credited through those later exact-enumeration sources; its full text was not read here. Robinson’s full 1976 paper was also not inspected. Generic asymptotic work on Stirling numbers, cited in Kotěšovec’s discussion, is prior art for the analytic method. No exhaustive survey of that literature is claimed. The report’s mathematical conclusions rest on the proofs above, while the absence of a symbolic amplitude or diagonal correction in inspected sources supports only this bounded description of the contribution.

Several questions remain useful.

1. Can one obtain practical certified constants and an explicit onset for the relative remainder and inverse enclosure? That requires quantitative global contour gaps and Taylor bounds, beyond diagnostic fitting.
2. What is the growth of $c_j^{(p)}$ as $J \rightarrow \infty$? The fixed-order proof gives no convergence or optimal-truncation theorem.
3. Is there a simpler closed description of the first parity-dependent contribution at every later order, avoiding full two-variable expansion?
4. How does the analysis change for $k = \alpha n + O(1)$ bishops or rectangular boards? The two-color exact structure suggests nearby saddle problems, but their uniformity and possible transitions require a separate proof.
5. Does a complete inspection of the older original literature change the historical scope of the symbolic amplitude or refinements? That question is independent of the correctness of the present argument.

[write] **Further questions added under the standing rule of 4 October 2026, and dated notes, 6 October 2026.** On Question 1: Proposition 2.2 makes the onset of monotonicity explicit ($n_0 = 1$), but the onsets of (2) and (32) remain existential. On Question 5: A002465 links Arshon’s paper (*Matematicheskoe prosveshchenie*, Ser. 1, 8 (1936), 24–29) on Math-Net.Ru and cites Ahrens (1921, Vol. 1, p. 271); neither was read at the write, and Kotěšovec’s book was not reachable at the address the source gives (Section 1.1). Two claims of the source are unproved as delivered and are recorded here as questions, with what is missing.

6. *An independent order-4 check.* The delivered README says that the order-4 receipt (`data/reference_receipts-order4-coefficients.json`) is angular-only, “with no independent order-4 affine comparison”. Its $c_4^{(0)} = 8.11568\dots$ and $c_4^{(1)} = 6.77647\dots$ agree with the write’s extrapolation of exact counts (8.1154, 6.7763; floating, uncertified). Missing: an independent exact computation of $c_4^{(p)}$, by the affine route or otherwise.
7. *Parity-branch enclosures.* The source says after Theorem 7.1 that separate parity-branch roots and parity-adjusted ceilings “offer an alternative”, and the code constructs formal branch centres x_p whose third coefficients differ by $-r^3/(4(r+1)A)$ (README of the code). No enclosure of the per-class inverses is proved. Sketch: run the proof of Theorem 7.1 on each parity class with the constant coefficients $\ell_j^{(p)}$ in place of (29), then apply the residue-class formula `p0:thm:staircase(4)` with $r = 2$ (Remark 7.2(c)). Missing: that proof written out, with its onset.

A The third correction in reusable form

Define

$$\begin{aligned} C_3(r) = & 135r^{20} - 540r^{19} - 3195r^{18} + 26572r^{17} - 21576r^{16} \\ & - 117156r^{15} + 235079r^{14} - 6816r^{13} - 669756r^{12} \\ & + 952564r^{11} + 1122759r^{10} - 2739096r^9 - 1945791r^8 \\ & + 4512672r^7 + 3004830r^6 - 4316220r^5 - 2844720r^4 \\ & + 1356480r^3 + 1257120r^2 + 570240r + 51840. \end{aligned}$$

Then

$$c_3^{(0)} = -\frac{rC_3(r)}{51840(r-1)^9(r+1)^6}, \quad c_3^{(1)} = c_3^{(0)} + \frac{r^3}{4(r+1)}.$$

This form is smaller than two separate degree-twenty numerators and explicitly preserves the proved parity difference. Both expanded numerators, and exact c_0, c_1, c_2 , are provided in the companion JSON data. The coefficients are rational functions in the symbol r ; the only transcendental step is selecting the real saddle root in (1) for evaluation.

References

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- [4] J. Perott, Sur le problème des fous, *Bulletin de la Société Mathématique de France* **11** (1883), 173–186. <https://www.numdam.org/item/10.24033/bsmf.267.pdf>.
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